higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

QUANTITY SURVEYING N5

(2050015)

29 November 2019 (X-paper)
09:00–13:00

REQUIREMENTS: **Dimension paper (OE 8/12)**
Abstract paper (OE 8/10)
Billing paper (OE 8/11)

Candidates may use their own unmarked Standard System Of Measuring Building Work.

Nonprogrammable calculators may be used.

This question paper consists of 5 pages and 2 addenda.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
QUANTITY SURVEYING N5
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. All the work in SECTION A must be done on the appropriate paper.
4. All the work in SECTION B must be done in the ANSWER BOOK.
5. Number the answers according to the numbering system used in this question paper.
6. Use only a blue or black pen.
7. Write neatly and legibly.

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SECTION A

On ADDENDUM A the plan view and sectional view of a dwelling are shown.

NOTE: Ignore roof measurements as questions are based on the specifications of the superstructure.

Masonry: Local stock bricks built in stretcher bond in 1:6 cement mortar
Bituminous SABS 211 type 2 DPC on walls
External semi-face brick

Carpentry and joinery: 19 mm × 76 mm meranti skirting
19 mm quadrant to skirting

Ceiling: 3 mm thick asbestos cement sheeting on a 38 × 38 mm SA pine brandering and H-section cover strips with 75 mm gypsum plaster cornice

Roof covering: Corrugated fibre-cement sheeting with matching ridge cap of 12 × 300 barge and fascia boards nailed to timber ends

Floor covering : PVC floor tiles

Concrete: 20 MPa

Plastering: 15 mm ICP 1:5 mix
12 mm cement screed 1:4 mix
12 mm granolithic 1:6 mix floor covering

QUESTION 1

1.1 Calculate the length of ALL the 220 mm thick brick walls. (3)

1.2 Calculate the length of ALL the 110 mm thick brick walls. (3)

1.3 Measure the item for the 220 mm thick brick walls in the superstructure. (2)

1.4 Measure the item for the 110 mm thick brick walls in the superstructure. (2)

1.5 Measure the item for the external facings. (2)
[12]

QUESTION 2

2.1 Measure the item for the concrete on the floor. (7)

2.2 Measure the item for the DPC. (3)

2.3 Measure the item for the external finishing of the walls. (4)

2.4 Measure the item for the internal finishing for the walls. (4)
[18]

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QUESTION 3

3.1 Measure ALL the items for the internal floor finish.

3.2 Measure the items for the granolithic and the quadrant.

3.3 Measure the items for the internal ceiling.

3.4 Measure the item for the cornice.

(4 × 5) **[20]**

QUESTION 4

You are required to present a bill for ALL the items measured in QUESTION 3.

4.1 Rewrite and do the squaring of the items of the ceiling. (5)

4.2 Abstract ALL the items of the ceiling only. (7)

4.3 Draw up the bills for the items of the ceiling. (8)

[20]

TOTAL SECTION A: 70

SECTION B

QUESTION 5

5.1 State the *trade* and the *unit of measurement* laid down by the Standard System on how the following should be measured:

5.1.1 Open-faced bulk excavations

5.1.2 Precast concrete lintels

(2 × 2) (4)

5.2 Explain the following terms:

5.2.1 Preambles

5.2.2 Prime cost

(2 × 3) (6)

[10]

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QUESTION 6: FINAL ACCOUNT

Prepare the final account and calculate the balance payment due to the contractor based on the relevant items in the summaries of the B.O.Q. given on ADDENDUM B with the following adjustments:

Prime cost and provisional sums adjusted to	R120 200,00
Net extra on variation orders to be	R11 200,76
Remeasurement of foundations	R90 100,20
Remeasurement of plumbing and drainage	R95 500,00
Remeasurement of site works	R130 720,00
Increased cost amounts to	R93 200,00
Previous payment to contractor amounts to	R1 005 110,00

[10]

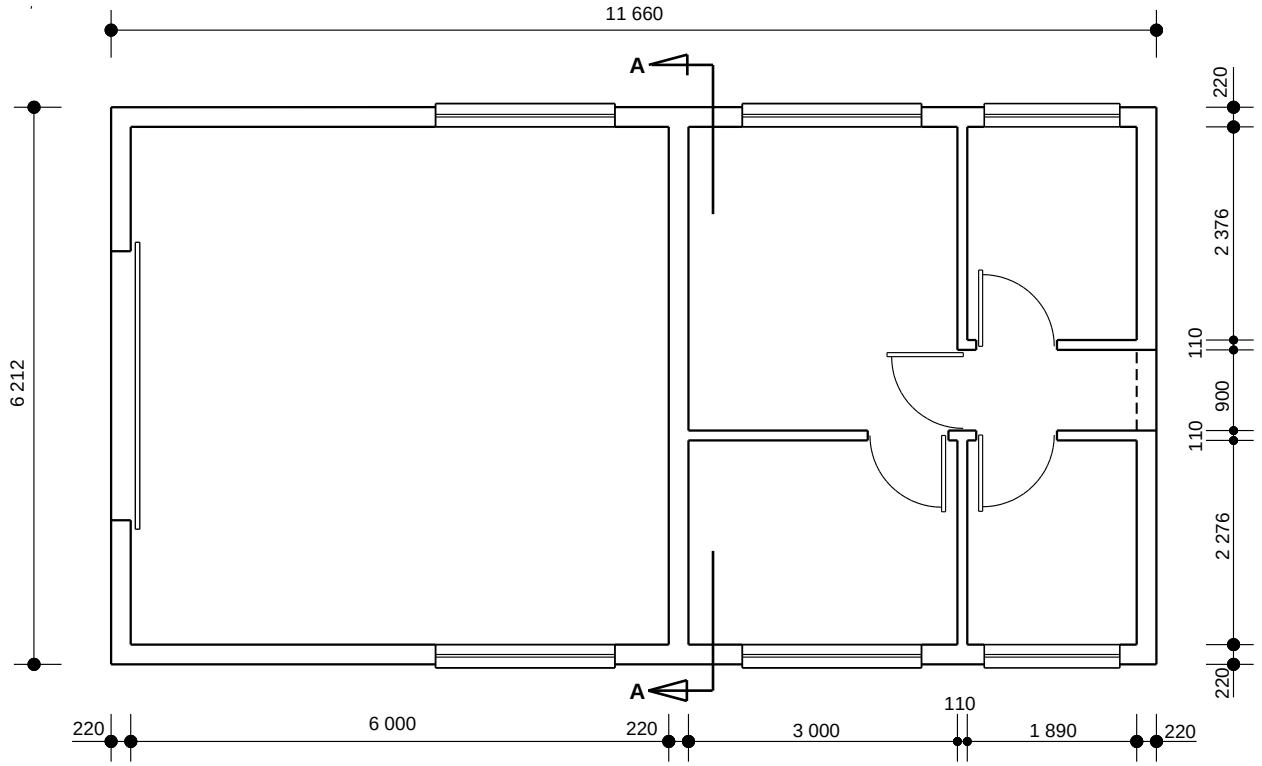
QUESTION 7

As quantity surveyor, name and explain the properties of concrete. **[10]**

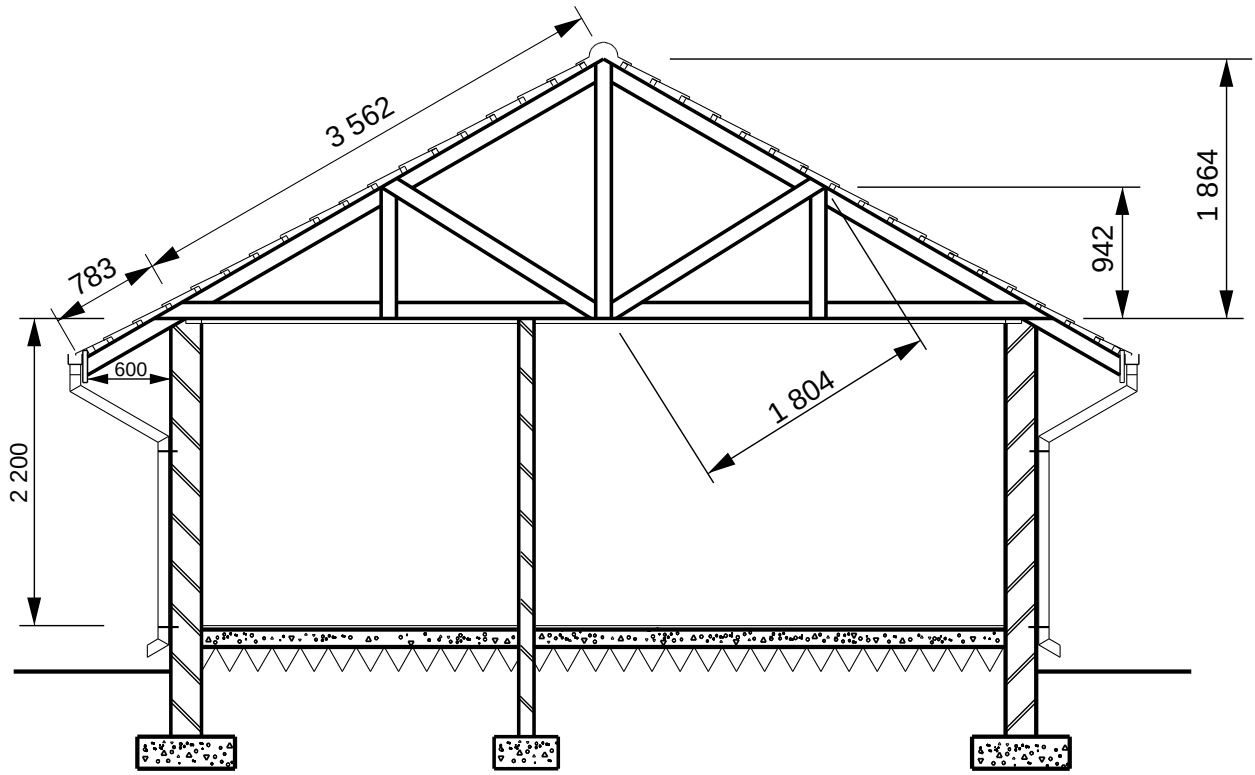
TOTAL SECTION B: 30
GRAND TOTAL: 100

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ADDENDUM A



PLAN VIEW



SECTIONAL VIEW

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ADDENDUM B

BILLS OF QUANTITIES

TRADE SUMMARIES

		AMOUNT	PAGE
1.	Foundations	R120 000,00	**80**
2.	Brickwork	R95 300,00	**90**
3.	Concrete, formwork and reinforcement	R250 200,00	**110**
4.	Masonry	R200 500,00	**99**
5.	Waterproofing	R20 310,00	**112**
6.	Roof covering	R150 220,00	**100**
7.	Carpentry and joinery	R65 300,00	**85**
8.	Ceilings	R25 300,00	**116**
9.	Floor coverings	R41 500,00	**120**
10.	Ironmongery	R8 520,00	**116**
11.	Metalwork	R25 000,00	**160**
12.	Plastering	R25 350,00	**101**
13.	Plumbing and drainage	R55 350,00	**175**
14.	Glazing	R24 300,00	**99**
15.	Site works	R135 300,00	**83**
16.	Prime cost and provisional sums	R162 155,00	**180**
17.	Contingency sum	R10 000,00	
		R13 244 605,00	

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